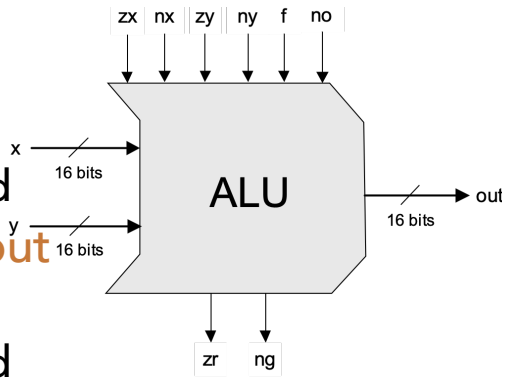
With a little help from my friends - implementing ALU

- Managing inputs
  - create logic circuits which, on the basis of zx, nx, zy, ny are able to select the proper 16-bit inputs
- Managing operators/outputs
  - create logic circuits which, on the basis of f, no are able to select the proper operation to be emitted
- Main idea
  - everything is computed altogether - the problem is selecting (i.e. MUXing)

With a little help from my friends - implementing ALU

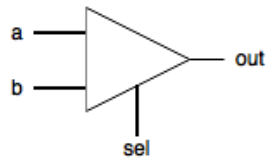
- As concerning the first input
  - it can be zero or x
  - it can be plain or bit-inverted
- As concerning the second input
  - it can be zero or y
  - it can be plain or bit-inverted
- As concerning operations and output
  - both ADD and AND must be evaluated
  - final output can be plain or bit-inverted
- As concerning the status bits
  - they codify some kind of semantic meanings



With a little help from my friends - implementing ALU

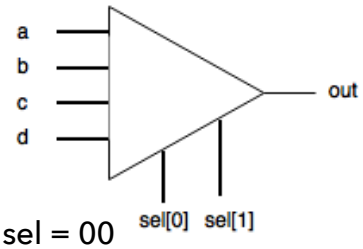
- Selecting with MUX

Selecting between 2 alternatives




```
IF sel = 0
      out = a
ELSE
      out = b
```

Selecting among 4 alternatives




```
IF sel = 00
      out = a
ELSEIF sel =01
      out = b
      …
ELSEIF sel =11
      out = d
```


With a little help from my friends - implementing ALU

- ALU implementation diagram 2

With a little help from my friends - tips for project 2

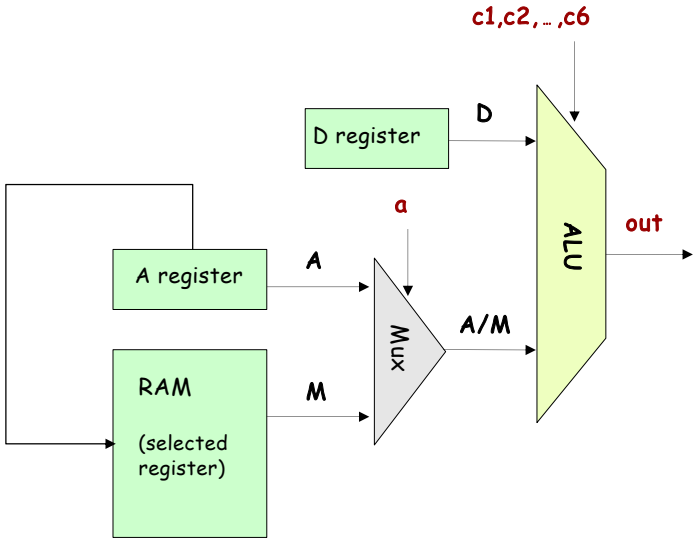
- For half adder two known gates can be used
- Full adder can be made up of 2 half adder (or in some other way)
- An n-bit adder can be built from n full adder chips
  - the carry bits are moved from right to left
  - the most significative-related carry is ignored
- For Inc16 remember that single "0", "1" bits can be expressed as "false", "true" in HDL
- ALU can be built with less than 20 parts
- Respect the given order and use built-in project-1 chips

With a little help from my friends - delivery instructions

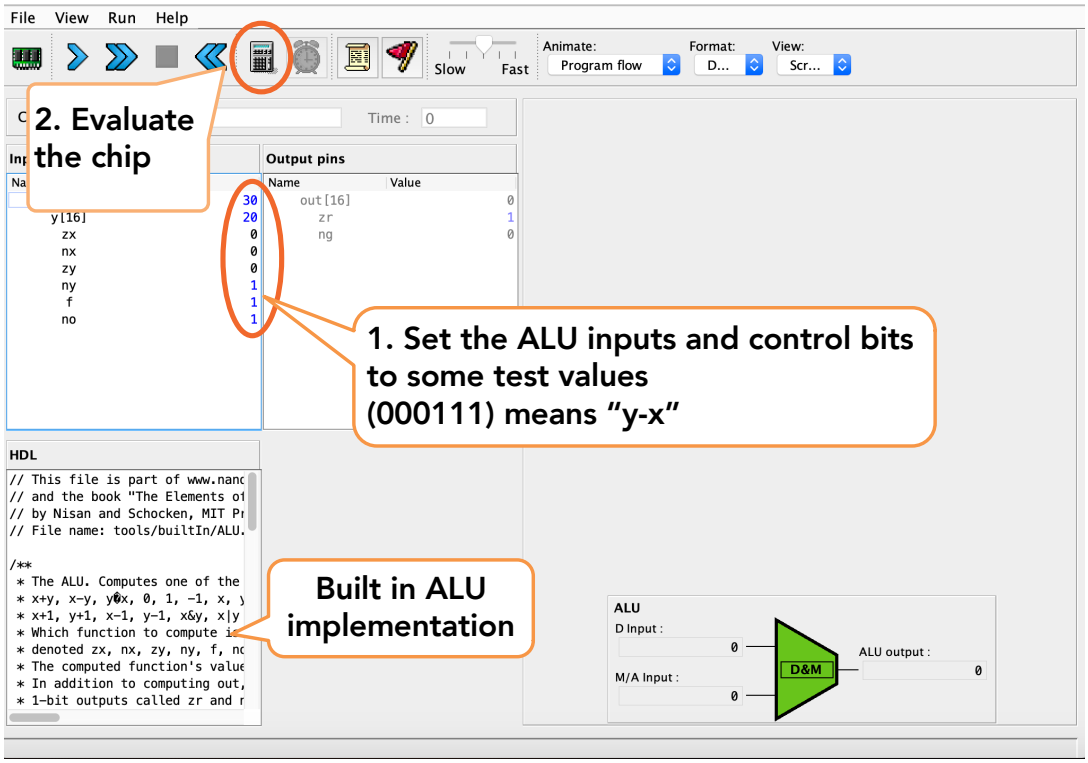
- Please adhere to the following notation for delivering the .hdl files
  - submit a zip file containing all (and only) the 5 .hdl files
  - pay attention to respect the following syntax for the name of the zip file

    _P02CognomeNome.zip (containing immediately 5 hdl files, with no additional directory)

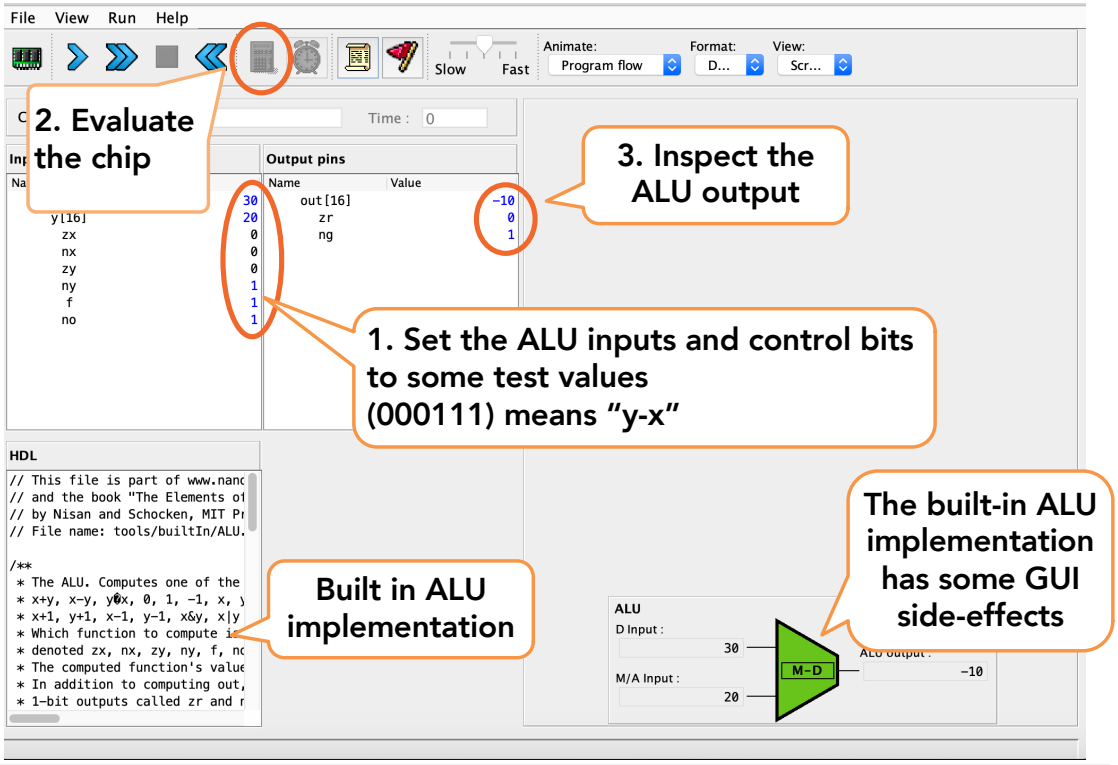
The ALU in the CPU context (a sneak preview of the Hack platform)

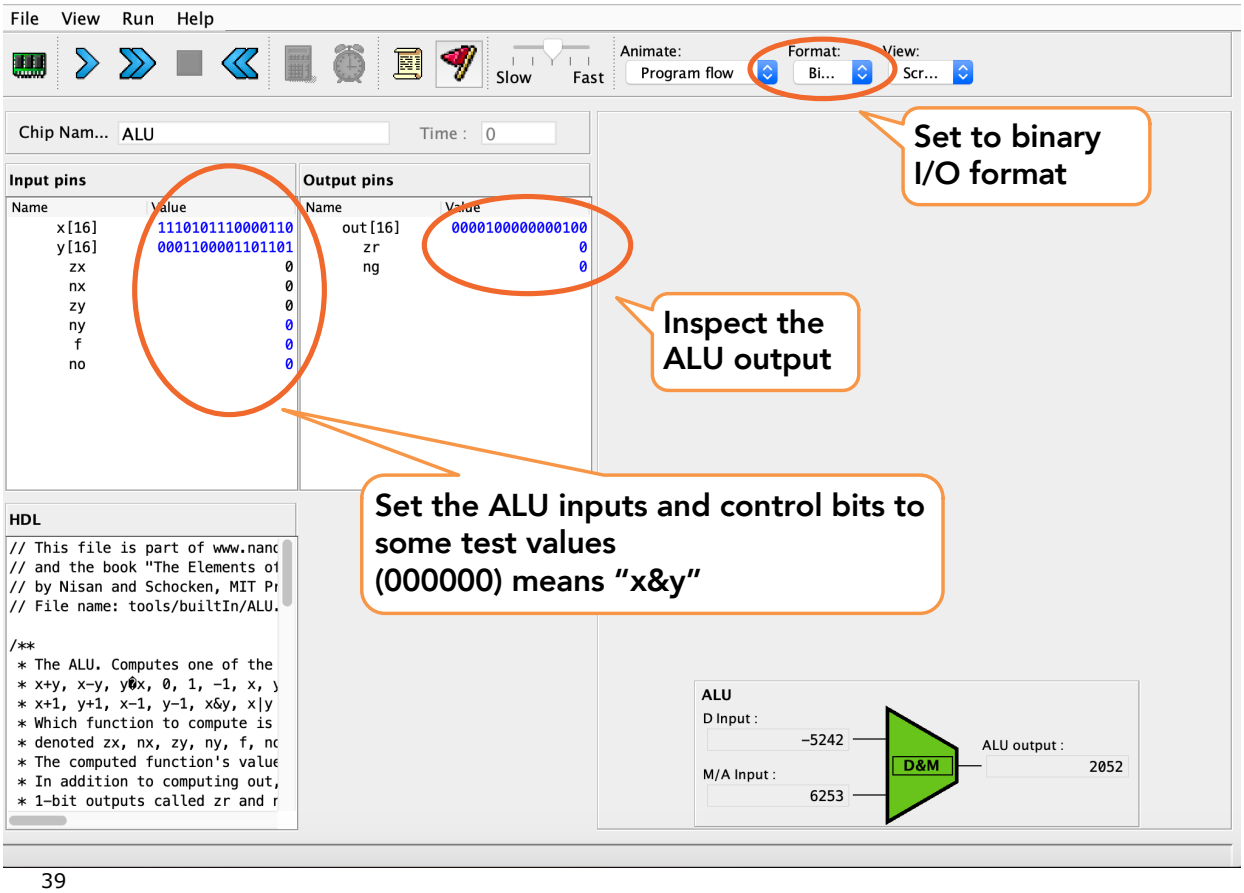


The ALU in action inside the hardware simulator



The ALU in action inside the hardware simulator





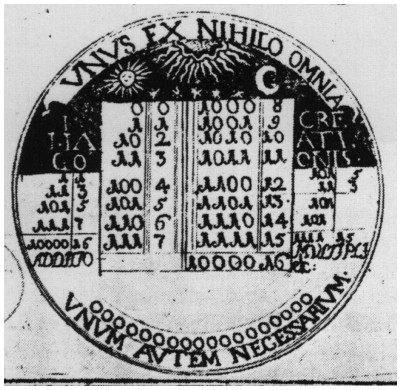
Perspective

- Combinational logic
- Our ALU is also very basic: no multiplication, no division
- Where is the seat of more advanced math operations?
  a typical hardware/software tradeoff.

Historical end-note: Leibnitz (1646-1716)

- "The binary system may be used in place of the decimal system; express all numbers by unity and by nothing"
- 1679: built a mechanical calculator (+, -, *, /)

- CHALLENGE: "All who are occupied with the reading or writing of scientific literature have assuredly very often felt the want of a common scientific language, and regretted the great loss of time and trouble caused by the multiplicity of languages employed in scientific literature:
- SOLUTION: "Characteristica Universalis": a universal, formal, and decidable language of reasoning
- The dream's end: Turing and Gödel in 1930's.



Leibniz's medallion
for the Duke of Brunswick